

## MATERIAL RATE DEPENDENCE AND MESH SENSITIVITY IN LOCALIZATION PROBLEMS

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The role of material rate dependence in setting the character of governing equations is illustrated in the context of a simple one-dimensional problem. For rate-dependent solids, the incremental equilibrium equations for quasi-static problems remain elliptic and wave speeds for dynamic problems remain real, even in the presence of strain-softening. The pathological mesh sensitivity associated with numerical solutions of localization problems for rate-independent solids is eliminated. In effect, material rate dependence implicitly introduces a length scale into the governing equations, although the constitutive description does not contain a parameter with the dimensions of length. Numerical results are presented that illustrate the localization behavior of slightly rate-dependent solids under both quasi-static and dynamic loading conditions.

### 1. Introduction

The localization of deformation into narrow bands of intense straining is a characteristic feature of inelastic deformation. Theoretically, for rate-independent solids localization is associated with a change in the character of the governing equations. Under quasi-static loading conditions the equations governing incremental equilibrium lose ellipticity, while under dynamic loading conditions wave speeds become imaginary [1–4]. Standard boundary value problems are then ill posed; one manifestation of this is that the width of the band of localized deformation is arbitrarily narrow. As a consequence numerical solutions to localization problems for rate-independent solids exhibit an inherent mesh dependence. The minimum width of the band of localized deformation is set by the mesh spacing. Furthermore, global quantities such as the overall stiffness characteristics of the body depend on the mesh size used to resolve the band of localized deformations. This has been observed in a variety of contexts and discussed from diverse perspectives, e.g. [5–8].

The inelastic response of materials is perhaps inevitably rate dependent. When material rate dependence is accounted for, there is no loss of ellipticity in quasi-static problems and wave speeds remain real, as long as stress levels remain small compared to *elastic* stiffnesses. Even though the phenomenology of localization can be the same for both rate-dependent and rate-independent material behavior (e.g. [9, 10]) pathological mesh dependence does not occur in numerical solutions for rate-dependent solids because boundary value problems remain well posed. The rate-independent solid does emerge as the appropriate limit of the rate-dependent solid, although the limiting case is a singular one. An abrupt change in

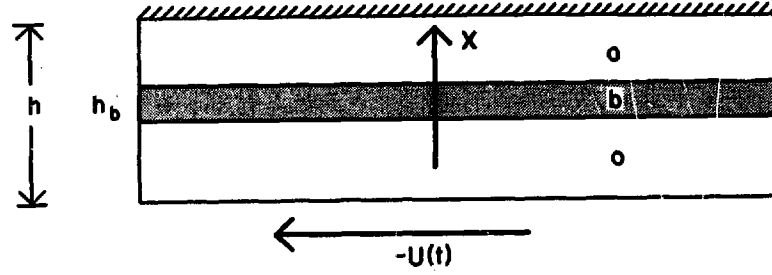


Fig. 1. Simple shearing of a planar strip. The only spatial dependence of field quantities is on  $x$ . Also shown is a planar band of thickness  $h_b$ . The relative thickness of the band is  $\beta = h_b/h$ .

behavior at the rate-independent limit also occurs in crystalline plasticity where there is a direct connection between rate dependence and uniqueness of slip mode [9, 11].

One aim here is to show, in the context of a simple one-dimensional problem, that material rate dependence eliminates the pathological mesh size effects encountered in rate-independent solids. In effect, material rate dependence introduces a length scale into the boundary value problem, even though the constitutive description does not contain a material parameter with the dimensions of length. For dynamic problems, this is implicit in the work of Wu and Freund [12], although their focus is on the mechanics of high strain-rate localization and their characterization of material rate dependence differs fundamentally from the one employed here. Another aim is to illustrate how the rate-independent limit is approached and, in particular, the sort of numerical problems that arise for slightly rate-dependent solids. For example, in the dynamic problem analyzed here, an ever increasing strain gradient is found.

The specific boundary value problem considered is sketched in Fig. 1; an infinite planar block subject to simple shearing displacement boundary conditions. Nonlinear kinematic effects that lead to possible normal strains or normal stresses are neglected. The character of the governing equations for rate-independent solids is discussed with a focus on factors that lead to the inherent mesh dependence accompanying localization. Then localization in rate-dependent solids is considered and numerical results are presented under both quasi-static and dynamic loading conditions.

## 2. Rate-independent solids

For a rate-independent elastic-plastic solid, the incremental shear stress-shear strain relation is written as

$$\dot{\tau} = G(\dot{\gamma} - \dot{\gamma}^p), \quad (2.1)$$

where  $G$  is the elastic shear stiffness,  $(\dot{\gamma})$  is  $\partial(\gamma)/\partial t$ , with  $t$  denoting time, and

$$\dot{\gamma}^p = \begin{cases} \dot{\tau}/H & \text{for plastic loading,} \\ 0 & \text{otherwise.} \end{cases} \quad (2.2)$$

Here,  $H$  is the plastic hardening modulus and plastic loading refers to circumstances where the

effective plastic strain,  $\bar{\gamma} = \int |\dot{\gamma}^p| dt$ , is increasing; i.e.  $\tau \geq \tau_y$  and  $\dot{\gamma}$  computed through (2.2) is positive.

Combining (2.1) and (2.2) gives

$$\dot{\tau} = \begin{cases} G_t \dot{\gamma} & \text{for plastic loading,} \\ G \dot{\gamma} & \text{otherwise,} \end{cases} \quad (2.3)$$

where  $1/G_t = 1/G + 1/H$ .

Although the focus here is on inelastic solids, the key feature for localization, the dependence of  $G_t$  on strain, does not depend on the material being characterized as inelastic. Localization becomes possible when  $G_t$  vanishes, whether the material is a rate-independent plastic solid, (2.3), or a nonlinear elastic solid ( $\dot{\tau} = G_t \dot{\gamma}$ , regardless of the strain increment direction).

### 2.1. Localization under quasi-static loading conditions

The boundary conditions for the simple shearing boundary value problem sketched in Fig. 1 are

$$v(0, t) = -\dot{U}(t), \quad v(h, t) = 0, \quad (2.4)$$

where  $v = \dot{u}$ , with  $u$  the displacement, and the sign in (2.4) is chosen so that positive  $\dot{U}(t)$  gives a positive shear rate.

The mesh dependence of numerical results in localization problems is an outcome of the character of the governing equations. Under quasi-static loading conditions, the change in character of the governing equations is associated with a bifurcation from a state of homogeneous shear into a deformation pattern with a shear band. The one-dimensional shear band bifurcation analysis is a specialization of the general three-dimensional analysis described by Rice [4]. Essentially this same one-dimensional model has been used in considering the necking of tensile bars [13], and localized buckling in compressed structures [14]. The presentation here, except for a change in notation, is taken from [14].

At some stage of the deformation history, the possibility of bifurcation into a shear band mode is considered, as sketched in Fig. 1. Equilibrium requires that the stress state remain homogeneous so that

$$\dot{\tau}_b = \dot{\tau}_o, \quad (2.5)$$

where  $( )_b$  and  $( )_o$  denote quantities associated with the corresponding regions in Fig. 1. With plastic loading in both regions, as required for the earliest bifurcation [15],

$$G_t(\dot{\gamma}_b - \dot{\gamma}_o) = 0. \quad (2.6)$$

A nontrivial solution to (2.6) is only possible when  $G_t$  vanishes so that here localization is associated with strain softening. However, it is important to emphasize that in a more general multi-axial context, strain softening is neither necessary nor sufficient for localization [4, 16].

In general, bifurcation into the shear band mode coincides with the governing incremental equations losing ellipticity [3, 4].

In the postbifurcation regime, the material inside the band continues to undergo plastic loading, while the material outside the band is elastically unloading (if the material outside the band were to continue plastic loading, the shear strain rates in the two regions would be identical and there would be no band). The total shear strain rate is

$$\dot{\gamma} = (1 - \beta)\dot{\gamma}_o + \beta\dot{\gamma}_b, \quad (2.7)$$

where  $\beta = h_b/h$ .

Using the equilibrium equation (2.5) and the constitutive relation (2.3) in (2.7) gives the overall shear stress-shear strain response as

$$\dot{\tau} = \frac{G_t \dot{\gamma}}{[\beta + (1 - \beta)G_t/G]}. \quad (2.8)$$

For a thick band,  $\beta$  near unity, the overall response follows the loading curve for the homogeneous material, while for a sufficiently thin band,  $\beta$  near zero, the overall response is close to the elastic unloading branch. Shear stress-shear strain curves illustrating the dependence on  $\beta$  are shown in Fig. 2. In this figure, the material is characterized by a piecewise linear hardening relation that will be used in subsequent numerical calculations for rate-dependent solids. (Of course, for piecewise linear hardening  $G_t$  is not defined at the peak. The curves in Fig. 2 are extrapolated back to the maximum shear stress as if  $G_t$  were zero there.)

The value of  $\beta$  is undetermined by the analysis and the solution is inherently nonunique. It is important to emphasize that the presence of an initial imperfection does not resolve this nonuniqueness. With an imperfection in the form of the band shown in Fig. 1, a maximum shear stress is attained in the band prior to the maximum shear stress being reached in the material outside the band. When the maximum shear stress is reached in the band, the

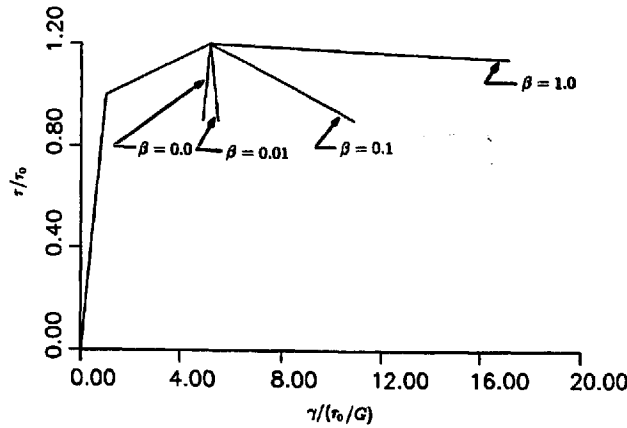


Fig. 2. The dependence of the overall shear stress-shear strain response on  $\beta$ , where  $\beta = h_b/h$ , for a rate-independent solid.

bifurcation arguments above apply except that the thickness,  $h$ , now refers to the thickness of the imperfection band. This is because the bifurcation condition (2.6) is a local one, which is also true in the full three-dimensional context [4]. Localization, i.e. loss of ellipticity, gives rise to inherently nonunique solutions with bifurcation modes in the form of arbitrarily narrow bands.

Numerical solution to such boundary value problems arbitrarily picks out one of the infinitely many possible solutions. The particular solution obtained is a consequence of the numerics since the solution is inherently nonunique. Typically the most unstable solution, i.e. the one with the narrowest band that the mesh can resolve, is the one seen in numerical solutions [5, 7].

## 2.2. Localization under dynamic loading conditions

Under dynamic loading conditions, the momentum balance and constitutive equation, (2.3), can be combined to give a wave equation for the velocity field,

$$\begin{cases} \rho v_{,tt} = G_t v_{,xx} & \text{for plastic loading} \\ \rho v_{,tt} = G v_{,xx} & \text{otherwise,} \end{cases} \quad (2.9)$$

where  $\rho$  is the mass density and commas denote partial differentiation.

For plastic loading (2.9) is a wave equation with wave speed  $c_t$  given by

$$c_t = \sqrt{G_t / \rho}. \quad (2.10)$$

A clear discussion of how a vanishing tangent modulus leads to “wave trapping” is given by Wu and Freund [12]. To summarize their discussion, suppose that at some shear strain  $G_t$  vanishes. Then the wave speed,  $c_t$  in (2.10), vanishes. In the particular boundary value problem under discussion here, the maximum strain occurs at the boundary  $x = 0$  so that  $c_t$  first vanishes at this point. With  $c_t = 0$ , the boundary itself is a characteristic and it is no longer possible to communicate boundary data to interior points. Additional increases in deformation are not transmitted to interior points and there is a strain discontinuity at  $x = 0$ .

Hence, under both quasi-static and dynamic loading conditions, the solutions to localization problems for rate-independent solids permit arbitrarily narrow bands of intense deformation. There is nothing in the formulation to set a minimum width to such bands. On the other hand, in numerical solutions the minimum width is set by the mesh spacing and key features of the solutions can be a consequence of the discretization.

This is clearly an undesirable state of affairs and stems from the character of the continuum problem. One approach to remedying this situation is to explicitly introduce a length scale into the material characterization by using a nonlocal constitutive relation [8]. Here, we will show that material rate dependence, no matter how small, leads to well-posed boundary value problems with unique solutions. This can be viewed in terms of material rate dependence implicitly introducing a length scale into the boundary value problem formulation. Of course, in any particular circumstance, the question remains as to whether or not this is the relevant length scale.

### 3. Rate-dependent solids

For a rate-dependent solid, the plastic strain rate is written as

$$\dot{\gamma}^p = \dot{a}f(\tau, g), \quad (3.1)$$

where  $\dot{a}$  is a material parameter of dimension 1/time and  $g$  is a hardness parameter that evolves with accumulating plastic strain. Also by writing (3.1) in terms of  $\dot{\gamma}^p$  and  $\tau$  rather than in terms of effective plastic strain rate and effective stress type quantities, attention is confined to circumstances where  $\tau$  is nonnegative. On quite general grounds, for metals undergoing plastic deformation due to dislocation motion, the plastic strain rate is appropriately taken as a function of the stress and the current state [25]. The relation (3.1) employs the idealization of a single parameter characterization of the current state. In the present context, the fundamental distinction between rate-dependent material behavior as embodied in (3.1) and the rate-independent behavior in (2.2) is that the plastic strain rate in (3.1) does not depend on incremental quantities.

Substituting (3.1) into (2.1) gives

$$\dot{\tau} = G[\dot{\gamma} - \dot{a}f(\tau, g)]. \quad (3.2)$$

In applications a power law representation of strain-rate dependence is often convenient and  $f(\tau, g)$  is written as

$$f(\tau, g) = (\tau/g)^{1/m}, \quad (3.3)$$

where  $m$  is the strain-rate hardening exponent. A power law description of material strain-rate sensitivity is not a fundamental physical relation, although it can provide a reasonable representation of the strain-rate sensitivity of common structural metals over several decades of strain rate, say from  $10^{-3}$  per second to  $10^3$  per second. The physical background for power law viscoplasticity, and its limitations, are discussed by Asaro [26] and Klopp, Clifton and Shawki [27]. For structural metals at room temperature, values of  $m$  in the range 0.002 to 0.02 are representative; see, for example, data collected by Ghosh [28] and also shown in [13, Fig. 1].

In the numerical examples,  $g$  is taken to be a function of the effective plastic strain  $\bar{\gamma}$  and a bilinear strain hardening function is used so that the evolution equation for  $g$  takes the form

$$\dot{g} = \begin{cases} H_1 \dot{\bar{\gamma}} & \text{for } \bar{\gamma} \leq \bar{\gamma}_m, \\ H_2 \dot{\bar{\gamma}} & \text{for } \bar{\gamma} > \bar{\gamma}_m, \end{cases} \quad (3.4)$$

with  $g(0) = \tau_0$  and  $g$  continuous at the transition strain  $\bar{\gamma}_m$ .

Figure 3 shows computed stress-strain curves for homogeneous shear at various imposed strain rates  $\dot{\gamma}/\dot{a} = |\dot{U}/h\dot{a}|$ , for  $\tau_0/G = 0.002$ ,  $m = 0.02$ ,  $\bar{\gamma}_m = 4\tau_0/G$  and  $H_1/\tau_0 = 25$ ,  $H_2/\tau_0 = -2.5$ . No explicit yielding is incorporated into the material description. Nevertheless, there is a fairly sharp transition at  $\tau = g(0)$ . Furthermore, under strain reversal the response virtually corresponds to what constitutes elastic unloading in the rate-independent theory. The curves

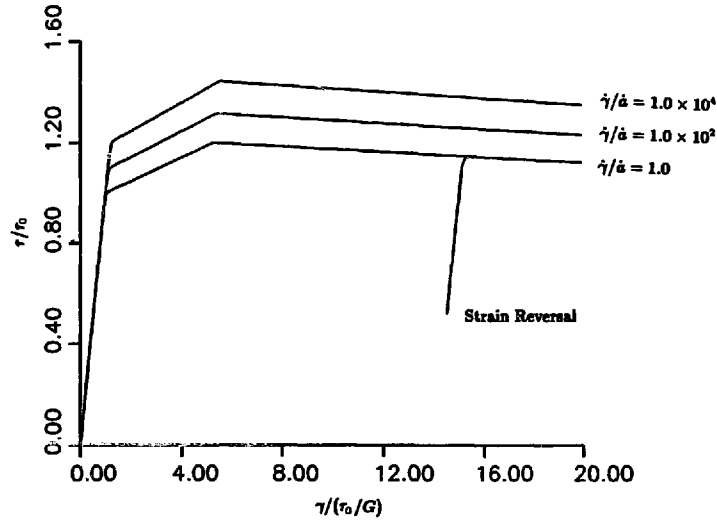


Fig. 3. Shear stress-shear strain response at various imposed strain rates for a rate-dependent solid with  $m = 0.02$ . Also shown is the “unloading”-like response on strain reversal, even though explicit elastic unloading is not incorporated into the constitutive description.

in Fig. 3 were obtained through a step-by-step integration of the constitutive equations, (3.2)–(3.4), using the rate tangent modulus scheme of Peirce, Shih and Needleman [17].

### 3.1. Localization under quasi-static loading conditions

The shear stress-shear strain curves in Fig. 3 are solutions to the quasi-static simple shearing problem. For the rate-dependent solid these solutions are unique and bifurcation into a shear band mode is ruled out. Substituting (3.2) into the rate equilibrium equation (2.5) gives the bifurcation condition

$$G(\dot{\gamma}_b - \dot{\gamma}_0) = 0. \quad (3.5)$$

Since the elastic shear modulus,  $G$ , is positive, the unique solution to (3.5) is  $\dot{\gamma}_b = \dot{\gamma}_0$ .

In fact, bifurcation into any mode is excluded. To show this the difference between two possible solutions is constructed. With  $\Delta(\ )$  denoting the difference between field quantities associated with the two possible solutions, rate equilibrium implies that the difference between any two solutions must satisfy

$$\int_0^h \Delta \dot{\tau} \Delta \dot{\gamma} \, dx = 0. \quad (3.6)$$

Using (3.2) in (3.6), together with the fact that  $f(\tau, g)$  is independent of rate quantities and thus is the same for both possible solutions, gives

$$\int_0^h G[\Delta \dot{\gamma}]^2 \, dx = 0. \quad (3.7)$$

The only solution of (3.7) with  $G > 0$  is  $\Delta\dot{\gamma} \equiv 0$ . Hence, the shear stress-shear strain curves in Fig. 3 give the unique solution for a homogeneous block subject to simple shear.

Next, suppose that there is a band in which the material properties differ from those in the surrounding material. A solution can be constructed in which there is uniform straining inside the band and uniform straining outside the band, but with a jump in strain rate across the band.

Within the band,

$$(\dot{\gamma}^p)_b = \dot{a}_b f_b(\tau, g_b), \quad (3.8)$$

and outside the band

$$(\dot{\gamma}^p)_o = \dot{a}_o f_o(\tau, g_o). \quad (3.9)$$

The rate equilibrium and compatibility equations are (2.5) and (2.7), respectively. Combining (3.8) and (3.9) with (2.5) and (2.7) gives two equations for the two unknowns,  $\dot{\gamma}_o$  and  $\dot{\gamma}_b$ ,

$$(1 - \beta)\dot{\gamma}_o + \beta\dot{\gamma}_b = \dot{\gamma}, \quad (3.10)$$

$$-\dot{\gamma}_o + \dot{\gamma}_b = -\dot{a}_o f_o(\tau, g_o) + \dot{a}_b f_b(\tau, g_b). \quad (3.11)$$

The determinant of coefficients of (3.10) and (3.11) is unity for all values of  $\beta$ . Hence, the band solution is unique and the imperfection sets the width of the band. Furthermore, an analysis paralleling (3.5) and (3.6) shows uniqueness with respect to other modes.

Figure 4(a) shows shear stress-shear strain curves for various imperfection amplitudes at an

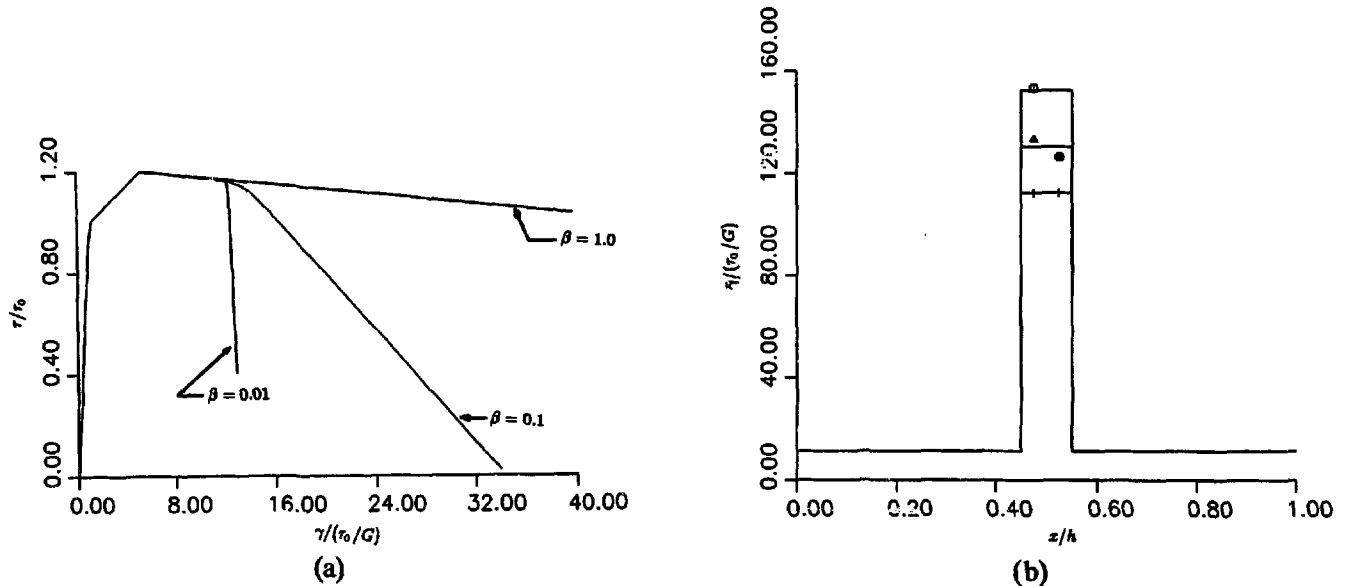


Fig. 4. (a) Overall quasi-static shear stress-shear strain response for a rate-dependent solid with  $m = 0.02$  and with imperfections having various widths  $\beta = h_b/h$ . The imposed strain rate is  $\dot{\gamma} = \dot{a}$ . (b) Strain distributions showing the development of numerical instability within the shear band for  $m = 0.02$  and  $\beta = 0.1$ . The numerically stable solution (solid lines) is obtained with  $\theta = 0$  in (3.16) and (3.17), while the points indicate the strain in each of the two elements in the band in a calculation with  $\theta = 1$ ; O:  $\tau/\tau_0 = 0.478$ ;  $\Delta$ :  $\tau/\tau_0 = 0.586$ ; and +:  $\tau/\tau_0 = 0.685$ .

imposed strain rate  $\dot{\gamma} = \dot{a}$ . Here,  $\dot{a}_0 = \dot{a}_b = \dot{a}$  and the parameter  $\dot{a}$  factors out of the governing equations by introducing the dimensionless time scale  $\bar{t} = \dot{a}t$ . The imperfection takes the form of a band in which  $\bar{\gamma}_m$  is 2.5% smaller than in the surrounding material. The calculation with  $\beta = 0.1$  is continued to nearly vanishing  $\tau$ . The amount of the delay in the onset of localization beyond the maximum stress point depends on the strain rate hardening [13] as well as on the imperfection. The post-localization response in Fig. 4(a) depends on the width of the imperfection band and the width of the imperfection band is the governing length scale in the problem. Much of the overall behavior is consistent with trends expected based on rate-independent material behavior. For example, once deformation begins to localize in the band, the material outside the band essentially ceases to deform. An important distinction is that for any given imperfection, the solution for rate-dependent material behavior is unique and the inherent mesh sensitivity of the rate-independent theory is eliminated. However, numerical stability problems do arise as the rate-independent limit is approached.

The solutions in Fig. 4(a) were obtained by a one-dimensional finite element analysis using linear displacement elements with a linear incremental time integration scheme based on the rate tangent modulus method of Peirce et al. [17]. The forward gradient expansion used in the calculations reveals the singular character of the rate-independent limit.

In forward gradient type methods, a linear interpolation for the plastic strain rate is used in each time increment,

$$\dot{\gamma}^p = (1 - \theta)\dot{\gamma}_i^p + \theta\dot{\gamma}_{i+\Delta t}^p, \quad (3.12)$$

where  $\theta$  is an interpolation parameter that is specified to lie between 0 and 1.

A Taylor-series expansion for  $\dot{\gamma}_{i+\Delta t}^p$ ,

$$\dot{\gamma}_{i+\Delta t}^p = \dot{\gamma}_i^p + \frac{\partial \dot{\gamma}^p}{\partial \tau} \Delta \tau + \frac{\partial \dot{\gamma}^p}{\partial g} \Delta g + \dots \quad (3.13)$$

is substituted into (3.12). Then, (3.12) and (3.13) in conjunction with (3.2) and (3.3) give

$$\dot{\tau} = G_{\tan} \dot{\gamma} - \dot{q}, \quad (3.14)$$

where

$$G_{\tan} = G \left[ 1 - \frac{\xi}{1 + \xi} \frac{1}{1 + (g'\tau)/(Gg)} \right], \quad \dot{q} = \frac{G\dot{\gamma}_i^p}{1 + \xi}, \quad (3.15)$$

with  $\dot{\gamma}_i^p$  given by (3.1),  $g'$  denoting the derivative of  $g$  with respect to its argument, and

$$\xi = \frac{\theta \Delta t}{m} \left[ \frac{G[1 + (g'\tau)/(Gg)]\dot{\gamma}_i^p}{\tau} \right]. \quad (3.16)$$

The modulus  $G_{\tan}$  in (3.17) is not a material property since its value depends on parameters of the time integration scheme as well as on constitutive parameters. The rate-independent limit is  $\xi \rightarrow \infty$  and with  $\xi = 0$  (3.16) reduces to (3.2). Now, in (3.15), if  $m$  is fixed at some nonzero value and the limit  $\theta \Delta t \rightarrow 0$  is taken,  $\xi \rightarrow 0$  so that  $G_{\tan} \rightarrow G$ . On the other hand, if  $\theta \Delta t$  has a fixed positive value, the rate-independent limit ( $G_{\tan} \rightarrow G$ ,  $\dot{q} \rightarrow 0$ ) is approached in (3.15) as  $m \rightarrow 0$ .

With  $\theta = 0$ , i.e. with Euler integration, numerical stability considerations limit the time step to be such that the strain increment is of the order  $m \times \tau_0/G$  [20]. Hence, the stable time step vanishes with  $m$ . For any positive  $m$ , sufficiently small steps in conjunction with  $\theta = 0$  give numerically stable time integration. By way of contrast, (3.15) and (3.16) show that with  $\theta \Delta t \neq 0$  and  $g'$  negative numerical stability is eventually lost. Since equilibrium requires  $\tau$  to have the same value inside the band as outside the band while  $g$  can vanish in the band, the quantity  $(g'\tau)/(Gg)$  appearing in (3.15) and (3.16) becomes large and negative at some point. The modulus  $G_{\text{tan}}$  in (3.15) then vanishes or becomes negative.

Figure 4(b) shows the development of the strain concentration in the band for  $\beta = 0.1$ , calculated using two values of  $\theta$ :  $\theta = 0$  and  $\theta = 1$ . In both calculations the spatial discretization has two elements in the band. The results of the two calculations are in very good agreement as long as numerical stability is maintained. With  $\theta = 0$  numerically stable results are obtained all the way down to  $\tau/\tau_0 = 0.02$ , while with  $\theta = 1$  a numerical instability occurs at  $\tau/\tau_0 = 0.64$ . As shown in Fig. 4(b), when the instability occurs the strain accumulation in the band is about  $30 \times \bar{\gamma}_m$ , with  $\bar{\gamma}_m = 4\tau_0/G$ . The numerical instability mode is identical to that accompanying the inherent instability in rate-independent solids. The deformation localizes into one element in the band and the shear stress then falls more rapidly, consistent with the narrower band. However, there is a fundamental distinction between the instability exhibited in Fig. 4(b) and that encountered in rate-independent plasticity; using  $\theta = 0$  and taking sufficiently small time steps, the calculations for the rate-dependent solid converge to the exact solution, which, furthermore, is unique.

While the details of the numerical instability clearly depend on the integration scheme used, (3.15) and (3.16) are consistent Taylor expansions for the rate-dependent solid, and the instability exhibited in Fig. 4(b) is undoubtedly characteristic for a broad class of forward gradient and implicit schemes. In this context, it is curious to note that if the spatial discretization has one element across the band, then the numerical instability in Fig. 4(b) is filtered out and an accurate solution is attained even when using  $\theta = 1$  and rather large (compared to those required by the Euler method) step sizes. It is also worth noting that material rate dependence eliminates inherent mesh sensitivity by giving rise to a unique solution and not by smoothing out the strain discontinuity across the band.

### 3.2. Localization under dynamic loading conditions

The character of the governing equations is revealed by differentiating the equation of motion

$$\rho u_{,tt} = \tau_{,x} \quad (3.17)$$

with respect to time and using the constitutive relation (3.2) to obtain

$$\rho v_{,tt} = Gv_{,xx} - G\dot{a}f(\tau, g)_{,x} \quad (3.18)$$

The material parameters in (3.18) can be grouped into a characteristic time,  $T$ , a characteristic length,  $L$ , via

$$T = \frac{1}{\dot{a}}, \quad L = \frac{\sqrt{G/\rho}}{\dot{a}} \quad (3.19)$$

The parameter  $L$  is a propagation length; it is the distance an elastic wave travels in the characteristic time  $T$ .

Introducing the dimensionless variables

$$\bar{t} = \frac{t}{T}, \quad \bar{x} = \frac{x}{L}, \quad \bar{v} = \left(\frac{T}{L}\right)v, \quad (3.20)$$

into (3.18) gives

$$\bar{v}_{,\bar{t}} = \bar{v}_{,\bar{x}\bar{x}} - f(\tau, g)_{,\bar{x}}. \quad (3.21)$$

The last term in (3.21) can be rewritten as

$$f(\tau, g)_{,\bar{x}} = f(\tau, g)_{,\tau}\tau_{,\bar{x}} + f(\tau, g)_{,g}g_{,\bar{x}} = Gf(\tau, g)_{,\tau}\bar{v}_{,\bar{t}} + f(\tau, g)_{,g}g_{,\bar{x}}. \quad (3.22)$$

Substituting (3.22) into (3.21) gives

$$\bar{v}_{,\bar{t}} + \lambda\bar{v}_{,\bar{t}} = \bar{v}_{,\bar{x}\bar{x}} + \xi\bar{g}_{,\bar{x}}, \quad (3.23)$$

where

$$\lambda = Gf(\tau, g)_{,\tau}, \quad \xi = -Gf(\tau, g)_{,g}, \quad \bar{g} = \frac{g}{G}. \quad (3.24)$$

For the power law relation (3.3)

$$\lambda = \frac{G(\tau/g)^{1/m}}{m\tau}, \quad \xi = \frac{\lambda\tau}{g}. \quad (3.25)$$

The governing equations are (3.23) together with the evolution equation for  $g$  given by (3.4). The highest-order derivative terms in (3.23) correspond to a wave equation with characteristics at  $45^\circ$  in the  $\bar{x}-\bar{t}$  plane. This directly leads to one key feature consistent with observation, incremental disturbances propagate with the elastic wave speed, see e.g. [21]. The character of (3.23), as determined from the highest-order derivative terms, is independent of whether the material is hardening or softening, which contrasts with the situation for rate-independent solids. Using the nondimensional variables (3.20) in (2.9), the character of the equation for plastic loading depends on whether the material is hardening or softening. If a characteristic length based on  $G_t$  is employed in (2.9), then the normalization breaks down when  $G_t$  vanishes. The rate-independent limit is  $m \rightarrow 0$ , which corresponds to  $\lambda \rightarrow \infty$  and  $\xi \rightarrow \lambda$  in (3.23). In this limit only the lower-order derivative terms remain in (3.23).

Numerical solutions were obtained directly from (3.17) based on one-dimensional linear displacement finite elements, using a lumped mass matrix. The time integration method is a central difference scheme that can be regarded as a member of the Newmark family of time stepping algorithms [19]. At each time increment the stress state is updated using the rate tangent method of Peirce et al. [17], although in most of the specific calculations here, the time steps used are so small that a simple Euler update of the constitutive relation gives almost identical results. The formulation using central difference time stepping with the rate tangent constitutive updating is described by Moran [18]. For the present problem, the basic equations take the form

$$\mathbf{u}^{n+1} = \mathbf{u}^n + \Delta t \mathbf{v}^n + \frac{1}{2}(\Delta t)^2 \mathbf{v}_{,t}^n, \quad (3.26)$$

$$\mathbf{v}_{,t}^{n+1} = -\mathbf{M}^{-1} \mathbf{F}^{n+1}, \quad (3.27)$$

$$\mathbf{v}^{n+1} = \mathbf{v}^n + \frac{1}{2} \Delta t [\mathbf{v}_{,t}^n + \mathbf{v}_{,t}^{n+1}], \quad (3.28)$$

where  $\mathbf{u}$  and  $\mathbf{v}$  are vectors of nodal displacement and velocity, respectively;  $\mathbf{M}^{-1}$  denotes the inverse of the mass matrix,  $\mathbf{F}$  is the vector of nodal forces, and superscripts  $n$  and  $n+1$  refer to quantities evaluated at the  $n$ th and  $(n+1)$ th time steps. At each integration point, the shear stress is updated via

$$\tau^{n+1} = \tau^n + \Delta t \dot{\tau}, \quad \dot{\tau} = G_{\tan} \frac{\Delta \gamma}{\Delta t} - \dot{q}. \quad (3.29)$$

Here,  $G_{\tan}$  and  $\dot{q}$  are given by (3.15) and  $\Delta \gamma = \gamma^{n+1} - \gamma^n$ , where the values of  $\gamma^{n+1}$  and  $\gamma^n$  are calculated from  $\mathbf{u}^{n+1}$  and  $\mathbf{u}^n$ , respectively.

Calculations were carried out for the simple shearing problem depicted in Fig. 1 with the boundary conditions (2.4) and with  $h$  equal to the characteristic length  $L$  in (3.19). The strain hardening and strain rate hardening characteristics are specified by (3.3) and (3.4), with the numerical values of material parameters as given in the text following (3.4), unless specifically stated otherwise. No inhomogeneity is prescribed.

The function  $\dot{U}(t)$  in (2.4) is taken as

$$\dot{U}(t) = \begin{cases} 2(\tau_0/G)(L/T)(t/t_1) & \text{for } t \leq t_1, \\ 2(\tau_0/G)(L/T) & \text{for } t > t_1, \end{cases} \quad (3.30)$$

with  $\tau_1 = T/5$ .

Figure 5 shows spatial profiles of normalized plastic strain,  $\bar{\gamma}/(\tau_0/G)$ , at various times. The solution consists of a fast elastic wave, with wave speed unity in the normalized variables (3.20), a slow plastic wave, and a layer near the surface in which  $\bar{\gamma} > \bar{\gamma}_m$ . The transition to softening occurs at  $\bar{\gamma}_m = 4(\tau_0/G)$ . If the width of the localization zone is defined by the distance over which  $\bar{\gamma}_m$  is exceeded, then clearly in Fig. 5, the localization zone increases in thickness as time increases. Nevertheless, the peak strain near the surface grows sufficiently rapidly so that the strain gradient at the boundary actually sharpens with increasing time.

This increasing strain gradient is a consequence of the strain softening. Figure 6 shows spatial profiles of normalized plastic strain for calculation where  $H_2/\tau_0 = +2.5$ , but all other material parameters are identical to those in Fig. 5. The peak strain on the surface does not exhibit the rapid increase that takes place when the material strain softens. Except for this thin surface layer the responses shown in Figs. 5 and 6 are nearly identical. For example, the penetration of the region with  $\bar{\gamma} > \bar{\gamma}_m$  is nearly the same in both cases. The results in Figs. 5 and 6 are each an outcome of (3.21); what differs in the two cases is the evolution of the forcing function  $f(\tau, g)$ .

Figure 7 shows spatial profiles of normalized plastic strain at a fixed time for three values of the strain rate hardening exponent  $m$ . The strain profile that develops is the outcome of two competing effects. The softening region diffuses into the material, while the strain peak at  $\bar{x} = 0$  increases. Increasing rate sensitivity delays the sharpening of the strain peak. However,

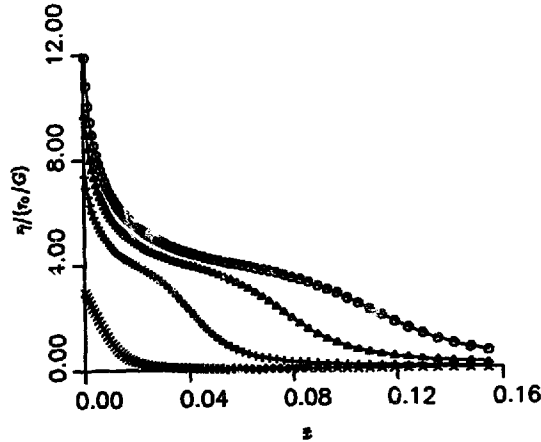


Fig. 5. Normalized plastic strain,  $\bar{\gamma}/(\tau_0/G)$ , versus position at various times for a softening solid,  $H_2/\tau_0 = -2.5$  in (3.4), with strain rate sensitivity  $m = 0.02$ ;  $\times$ :  $t/T = 0.170$ ;  $+$ :  $t/T = 0.340$ ;  $\Delta$ :  $t/T = 0.510$ ; and  $\circ$ :  $t/T = 0.680$ .

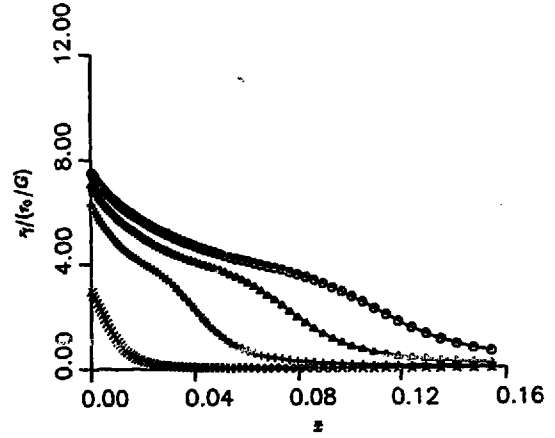


Fig. 6. Normalized plastic strain,  $\bar{\gamma}/(\tau_0/G)$ , versus position at various times for a hardening solid,  $H_2/\tau_0 = +2.5$  in (3.4), with strain rate sensitivity  $m = 0.02$ ;  $\times$ :  $t/T = 0.170$ ;  $+$ :  $t/T = 0.340$ ;  $\Delta$ :  $t/T = 0.510$ ; and  $\circ$ :  $t/T = 0.680$ .

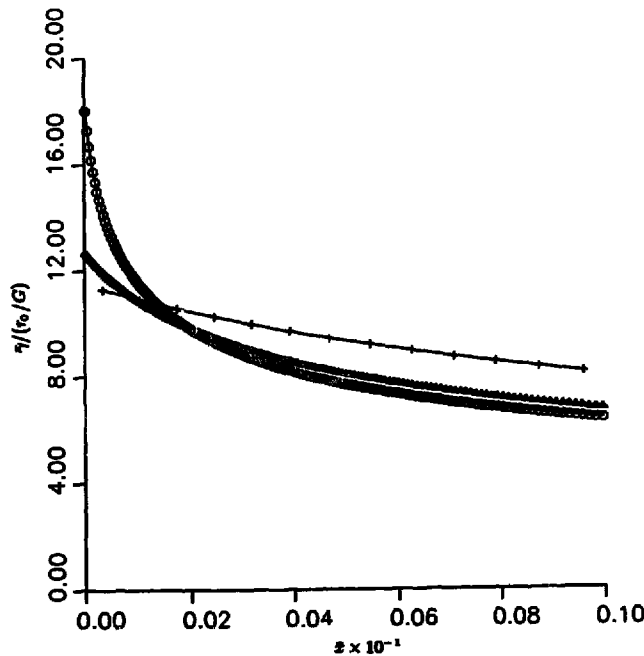


Fig. 7. Normalized plastic strain,  $\bar{\gamma}/(\tau_0/G)$ , versus position at  $t/T = 0.680$ ;  $+$ :  $m = 0.05$ ;  $\Delta$ :  $m = 0.02$ ; and  $\circ$ :  $m = 0.0152$ .

the spatial profiles for  $m = 0.05$  and  $m = 0.02$  continue to sharpen with increasing time and eventually take on the form shown in Fig. 7 for  $m = 0.0152$ .

For a very viscous solid,  $m = 0.5$ , calculations were carried out until the strain is attained at which the hardness  $g$  in (3.2) to (3.4) first vanishes. Three meshes, with 25, 50, and 100 elements, were used. As in the previous calculations, these meshes are graduated so that the element size is smallest near  $\bar{x} = 0$ , but the graduation is the same in each case so that the size of the element at  $\bar{x} = 0$  does vary by a factor of four in going from the coarse mesh to the fine

mesh. The calculations remain stable all the way until  $\tau$  vanishes in the element at  $\bar{x} = 0$ . This occurs at  $t/T = 28.1$ , 27.2, and 26.8 for the 25, 50, and 100 element meshes, respectively. These times are associated with plastic strain accumulations in the element where  $\tau$  vanishes of  $242\tau_0/G$  for the 25-point mesh and  $245\tau_0/G$  for the 50- and 100-point meshes. Hence, in this case there is convergence over the entire deformation history all the way to first "failure", i.e. when  $\tau$  in the most highly strained element vanishes. With regard to "failure", it is worth noting that, although the problem formulation here does not account for the work associated with the creation of new free surface, for ductile solids, the main contribution to the applied energy absorbed to failure is the inelastic dissipation. For a more nearly rate-independent solid, the computation is much more difficult, since strain gradients become very much sharper.

Figure 8 shows spatial profiles of normalized plastic strain for  $m = 0.0152$ , using three meshes. The solid line is the same curve as in Fig. 7, but the individual integration points are not marked. The coarsest mesh corresponds to the one used to obtain the results for  $m = 0.02$  shown in Fig. 5. The numerical problem is one often encountered in boundary layer problems. Convergence takes place in that the coarse mesh values fall right on the fine mesh results. However, in order to obtain an accurate estimate of the peak plastic strain, the mesh must be fine enough to resolve the steep gradients near  $\bar{x} = 0$ . The last three points on each of the curves in Fig. 8 was used to estimate the normalized plastic strain value at  $\bar{x} = 0$  by quadratic extrapolation. The values obtained are  $\bar{\gamma}/(\tau_0/G) = 16.0$ , 17.0, and 18.4 with increasing resolution. The numerical problem here is the difficult one of resolving a continually sharpening gradient, but convergence does appear to occur in these calculations with increasing mesh refinement.

The overall development of localization shown in Figs. 5, 7, and 8, i.e. a softening region

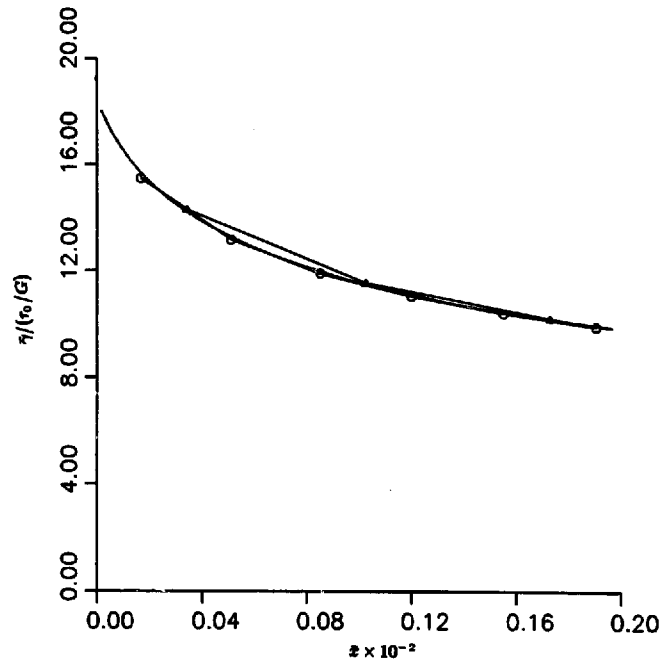


Fig. 8. Normalized plastic strain,  $\bar{\gamma}/(\tau_0/G)$ , versus position for various degrees of mesh refinement at  $t/T = 0.680$  for  $m = 0.0152$ . The solid line is reproduced from Fig. 7;  $\circ$  using a 200-point mesh,  $\Delta$  using a 100-point mesh.

that diffuses into the material in conjunction with a rapidly increasing peak strain, is much the same as in Wu and Freund [12] even though the constitutive relation they use leads to a governing equation of a different character from (3.21).

In the present notation their constitutive equation can be written as

$$\tau = F(\gamma) + \hat{\tau} \nu \dot{\gamma} . \quad (3.31)$$

Substituting (3.31) into (3.17) and using nondimensional variables as in (3.19), but with the characteristic time now being  $\nu$ , leads to a governing equation of the form

$$\bar{v}_{,\bar{t}} = \bar{v}_{,\bar{x}\bar{x}} + \bar{F}(\bar{\gamma})_{,\bar{x}} . \quad (3.32)$$

The form of the highest derivative terms in (3.32) is that of a diffusion equation whereas the highest derivative terms in (3.23) correspond to a wave equation. In both cases there is a coupled evolution equation for the inhomogeneous source type term.

#### 4. Concluding remarks

When material rate dependence is accounted for, pathological mesh size effects do not occur. Under quasi-static loading conditions, the incremental equilibrium equations are elliptic and, furthermore, solutions are unique. Under dynamic loading conditions wave speeds remain real. In the simple shear problem considered here, this is true regardless of the stress level. In more general circumstances, there is the requirement that stress levels remain small relative to elastic stiffnesses [4]. It should also be emphasized that, while in the simple shear problem considered here localization is associated with strain softening, in general strain softening is neither necessary nor sufficient for localization [4, 16].

In effect, material rate dependence introduces a level scale, illustrating that it is not necessary to introduce a characteristic length into the constitutive relation in order for a characteristic length to appear in the governing boundary value problem. In quasi-static problems, the length scale is one characterizing the imperfection or inhomogeneity; in dynamic problems, it is a characteristic length of propagation of elastic waves. In any particular case other length scales may be relevant and may in fact play a dominant role in setting the width of shear bands. Boundary value problems can be formulated incorporating the length scale that sets the shear band width; for example, by incorporating appropriate geometric or material inhomogeneities into the formulation. Also, accounting for heat conduction introduces a material-dependent length scale over which heat conduction effects are significant, e.g. [22]. In other circumstances the dimensions of some microstructural feature, e.g. a grain size, of a size scale smaller than any modelled in the boundary value problem formulation, may set the length scale. In such a case a constitutive characterization explicitly incorporating a material-dependent length parameter appears appropriate.

The numerical method used here for quasi-static problems clearly exhibits the singular nature of the rate-independent limit. With multiple elements in the band, a numerical instability is encountered, although for any finite amount of material rate sensitivity and using sufficiently small time steps, the calculations can remain stable out to large accumulations of

deformation in the shear band. When the band consists of a single element, accurate solutions can be obtained even with relatively large time steps, since the coarseness of the mesh filters out the numerical instability mode. The numerical instability reflects the actual tendency for shear bands to be as narrow as possible. In this vein, numerical localization results for rate-independent solids can be interpreted as corresponding to solutions for the limit of vanishing rate dependence for a solid with an imperfection, of whatever origin, that induces a shear band one element wide. Provided that the element size is such that this shear band width is appropriate for the particular situation being modelled, the numerically obtained solution is a pertinent one.

More severe numerical difficulties are associated with the dynamic localization problem. Although the governing equation has real wave speeds, there is an ever increasing strain gradient to resolve. The width of the region of high gradients depends on the material strain rate sensitivity and is narrower for less rate sensitive materials. As can be seen in Figs. 5, 7, and 8, the width of this region is a small fraction of the material-dependent length  $L$  defined in (3.20). In order to efficiently resolve the narrow shear bands that develop for slightly rate-independent solids, an adaptive mesh method, e.g. as has been used by Drew and Flaherty [23] in a shear localization problem, is clearly useful. Narrowing shear bands are observed [29]. In actuality, of course, under the extreme conditions of strain and strain rate that prevail in such a narrow shear band, material behavior is not described by the sort of simple phenomenological constitutive relation used in this investigation.

The problem considered here is a simple one-dimensional one. In multi-axial situations, important computational issues arise concerning the orientation of finite element meshes [5, 24], that do not arise in the one-dimensional context. The ability of a finite element mesh to resolve localized shearing at angles oblique to the element boundaries significantly affects the predicted course of shear band development. This sort of mesh dependence is an entirely different issue from the one considered here and arises whether or not the material is characterized as rate-independent or rate-dependent or, indeed, whether or not the constitutive relation incorporates a material-dependent length parameter.

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